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Exploring the effects of Game-based activities on spatial-temporal reasoning in Children aged 6 to 11 years

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Abstract :

This study examines the effect of a game-based intervention, specifically tangram, on improving spatial-temporal reasoning in children aged 6 to 11 years. Academic endeavors, diverse cognitive tasks, practical situations all necessitate spatial-temporal reasoning. Tangram is a traditional Chinese puzzle that enhances problem-solving abilities and spatial awareness. The study primarily aims to focus on the age group of 6 to 11 years, utilizing various theories of cognitive development. Children were randomly assigned to experimental and control groups utilizing an experimental design. A total of eighty students participated in the study. Speed and accuracy across three tasks were utilized to compare pre-test & post-test assessments of spatial-temporal reasoning. The experimental group's performance markedly enhanced post-intervention, as the results indicate the efficacy of the tangram puzzle. The study revealed that the intervention produced enduring effects, as participants demonstrated sustained improvements on the retention test conducted at 10-day intervals. The results illustrate how a tangram puzzle and an intervention can enhance children's cognitive development, impacting educational methodologies, and policy formulation.

Keywords : Tangram, spatial-temporal reasoning, game-based intervention, children aged 6 to 11 Preface

Introduction :

Spatial-temporal reasoning is the ability to understand three-dimensional relationships among objects in space and to cognitively manipulate them as a sequence of changes over time. (Newcombe and Shipley, 2015; Uttal, Meadow, et al., 2013). Spatial-temporal reasoning is essential for daily activities, such as maneuvering through space by navigating routes and anticipating turns while traveling, in addition to supporting fields like mathematics, and engineering, architecture. It assists children in perceiving visual artworks and facilitates the formation of mental images while enabling the recognition of various physical object variations. To enhance academic performance and cognitive development, it is essential to cultivate children's spatial-temporal reasoning skills. Cognitive training through games has been demonstrated to significantly enhance children's cognitive & spatial abilities

(Kim, S, et al., 2021; Uttal & Meadow, et al., 2013). A tangram is a quintessential puzzle consisting of seven flat polygons, commonly referred to as "tans." The polygons are organized to create diverse shapes in relation to a cue card. The primary aim of the game is to reproduce a design based on a cue card utilizing all seven pieces without any overlap. The Tangram originated from the 3rd-century Chinese mathematician Liu Hui. Tangrams were originally employed to instruct various mathematical concepts, including congruence, and symmetry, geometry, and perimeter. Subsequently, it was modified to instruct perceptual reasoning, visual-spatial awareness, and creativity. Studies demonstrate that Tangrams enhance children's observational skills, creative thinking, logical reasoning, and imagination (Yang and Chen, 2010; Olkun et al., 2005; Russell and Bologna, 1982).

Consequently, employing tangrams as a game can facilitate advancements in geometric skills, and including shape recognition, orientation identification, & the establishment of relationships among various two-dimensional geometric figures. Bohning and Althouse (1997); Krieger (1991). These experiences are essential for recognizing and valuing geometry in its natural context (Bohning and Althouse, 1997).

Background:

Cognitive abilities continue to develop during middle and late childhood. Processing concrete information results in more organized and logical thought processes. Between the ages of 6 and 7, children acquire concrete operational skills, including conservation. At ages 8 to 9, children can formulate and pursue objectives due to their comprehension of temporal concepts such as past, present, and future. They can also process complex concepts such as addition, subtraction, and cause-and-effect relationships. Theorists assert that children aged 10 to 12 engage in the generalization of concrete objects and events, representing the apex of concrete operations (Biggs and Collis, 1982). Some perceive it as the onset of the ability for hypothetical or abstract reasoning (Richards and Commons, 1983; Selman, 1980; Case, 1980; Fischer, 1980; Halford and Wilson, 1980; Jacques et al., 1978). The development of spatial-temporal thinking commences in early childhood and continues until puberty. During this critical period, children progressively acquire the ability to understand and anticipate the movement of objects in space and their transformations over time. (Jirout & Newcombe, 2015; Verdine, Golinkoff et al., 2014).

The extent to which this ability can be enhanced through systematic treatments remains a subject of ongoing research. Games have been employed to enhance cognitive functions and various health-related outcomes. According to Ferreira-Brito et al. (2019), employing Tangram is an effective method for enhancing children's spatial-temporal reasoning skills. Tangrams serve as effective cognitive training tools due to their engaging and participatory nature. Theorists such as Piaget assert that children aged 6 to 11 years are at a pivotal stage of cognitive development. Engaging in tangram activities is effective in enhancing spatial-temporal reasoning abilities in these children. Numerous studies have examined the impact of children's gameplay on their spatial and temporal reasoning abilities. Further research is necessary to thoroughly understand the potential benefits of games in enhancing children's spatial-temporal cognition. The importance of this study resides in its capacity to facilitate the use of games designed to enhance children's spatial-temporal reasoning. The research will illuminate the efficacy of games in enhancing spatial-temporal reasoning

in children and facilitate the development of tailored game-based interventions to meet individual needs. The research initiative will enhance understanding of the influence of spatial-temporal thinking on cognitive development and academic success. The research has the potential to facilitate the development of effective strategies to promote the enhancement of children's cognitive and spatial abilities.

Review of Literature:

Spatial-temporal reasoning includes abilities such as mental rotation, spatial visualization, and the utilization of temporal sequences, which are essential for numerous cognitive tasks. The investigation of game-based learning as a method to enhance cognitive ability is still relatively limited, despite its importance. Moreover, studies aimed at improving cognitive abilities in children aged 6 to 11 years are significantly scarce, particularly within the Indian context. Nevertheless, several studies have been conducted to comprehend spatial-temporal reasoning, which establishes a foundation for this research. Moreover, limited research has explored the impact of gaming on the enhancement of certain cognitive skills, illuminating potential pathways for improving cognitive abilities such as spatial-temporal reasoning through game-based activities in children. Rauscher et al. (1997) conducted a study to investigate the impact of music training on the enhancement of spatial-temporal reasoning in preschool children. The keyboard group demonstrated a notable enhancement in the Object assembly task, with 71% of participants improving their scores by 3-4 points.

A 2008 study by Beth M. Casey investigated the efficacy of block-building activities in enhancing kindergarteners' spatial reasoning skills and concluded that such activities could effectively aid young children in developing their spatial thinking abilities. Casey et al. (2008). In 2016, a study aimed to create a smart learning environment to enhance student engagement and motivation through game-based learning and gamification elements in medical contexts. The findings indicated that the implementation of game-based learning enhanced user engagement in content acquisition. Ferreira-Brito et al. (2019) sought to identify the game elements utilized in game-based interventions for cognitive assessment, training, or rehabilitation. It identified the most effective game elements for specific clinical conditions and cognitive outcomes. A study conducted by Lin, C., colleagues. In 2011, al. sought to create a virtual tangram to improve children's geometry learning within a computer-supported collaborative learning environment, demonstrating its effectiveness in enhancing spatial reasoning among children. A pilot study was performed to assess the effect of tangram on enhancing cognitive functioning in individuals with neurocognitive disorders

(Zhang, J. et al., 2023). The results indicated an enhancement in the cognitive functioning of patients in Study 1 following the tangram intervention. The literature demonstrates the considerable potential of game-based interventions in improving spatial-temporal reasoning abilities in children. Numerous studies substantiate the necessity of spatial-temporal reasoning & its adverse effects on daily functioning. Many reviewed papers exhibited a lack of cultural diversity, predominantly concentrating on Western populations, thereby impacting the generalizability of the findings. Moreover, certain studies include a wide age spectrum, potentially obscuring numerous age-specific effects and developmental subtleties. The age range of 6 to 11 is pivotal for cognitive development; however, there is a paucity of research focused specifically on this demographic. Understanding the nuances of spatial-temporal thinking development at this stage is essential. Consequently, the present study focuses on the 6 to 11 age demographic to obtain more accurate insights.

Methods:

Research Question:

Does engaging in tangram activities influence the development of spatial-temporal reasoning in children aged 6 to 11 years?

Aim:

To assess the influence of tangram play on the improvement of spatial-temporal reasoning in children aged 6 to 11 years.

Objectives:

1. To evaluate the efficacy of tangram as an intervention for improving spatial-temporal reasoning in children aged 6 to 11 years.
2. To evaluate the impact of Tangram play on the enhancement of spatial-temporal reasoning abilities over time, specifically examining the retention of cognitive improvements post-intervention.

Hypotheses:

H01a): There exists no significant disparity between the Pre-test control group and the Pre-test experimental group for tasks 1, 2, and 3.

H01b): There is no significant difference between the post-test control group and the post-test experimental group for tasks 1, 2, and 3.

H01c): There is no statistically significant difference between the pre-test experimental group and the post-test experimental group

H01d): There is no substantial difference between the pre-test control and post-test control for tasks 1, 2, and 3.

H02: There is no statistically significant difference among the Pre-test Experimental group, the Post-test Experimental group, and the Retention Test for tasks 1, 2, and 3.

Operational Definition:

Game: A game is an activity characterized by established rules, objectives, and challenges designed to entertain and engage participants. Games may be played solo or in groups, and they can be appreciated for recreational or educational purposes.

Tangram: It is a Chinese geometric puzzle comprising seven flat polygons that can be configured into various shapes, including animals and birds. Limited research indicates that tangrams enhance spatial-temporal reasoning as children learn to manipulate tans, facilitating mental rotation, spatial transformation, and pattern recognition. Therefore, it can serve as an intervention to augment cognitive capabilities. This intervention is readily accessible and straightforward to execute.

Spatial-temporal reasoning: Spatial-temporal thinking refers to the ability to cognitively modify spatial configurations through a sequential series of transformations over time. It involves understanding time and space, along with the mental representation and reasoning of spatial-temporal information.

Accuracy: Accuracy is defined as the extent to which the output aligns with or resembles the measured or observed value in relation to the actual or target value.

Speed: Speed is the ability to execute a task rapidly and accurately. It can be enhanced through cognitive training and various therapies and is commonly employed to assess cognitive performance.

Research design:

This study employs an experimental design. This study involved the random selection of subjects, who were assigned to either the experimental group or the control group. The participants were selected using convenience sampling. In the experimental group, students were instructed to complete three tasks, with speed and accuracy

assessed. Subsequent to task completion, the intervention game was administered. Following the intervention game, the students were instructed to repeat the tasks. The control group completed the three tasks and subsequently received a 15-minute break. The pre-test and post-test scores were assessed for both the experimental and control groups.

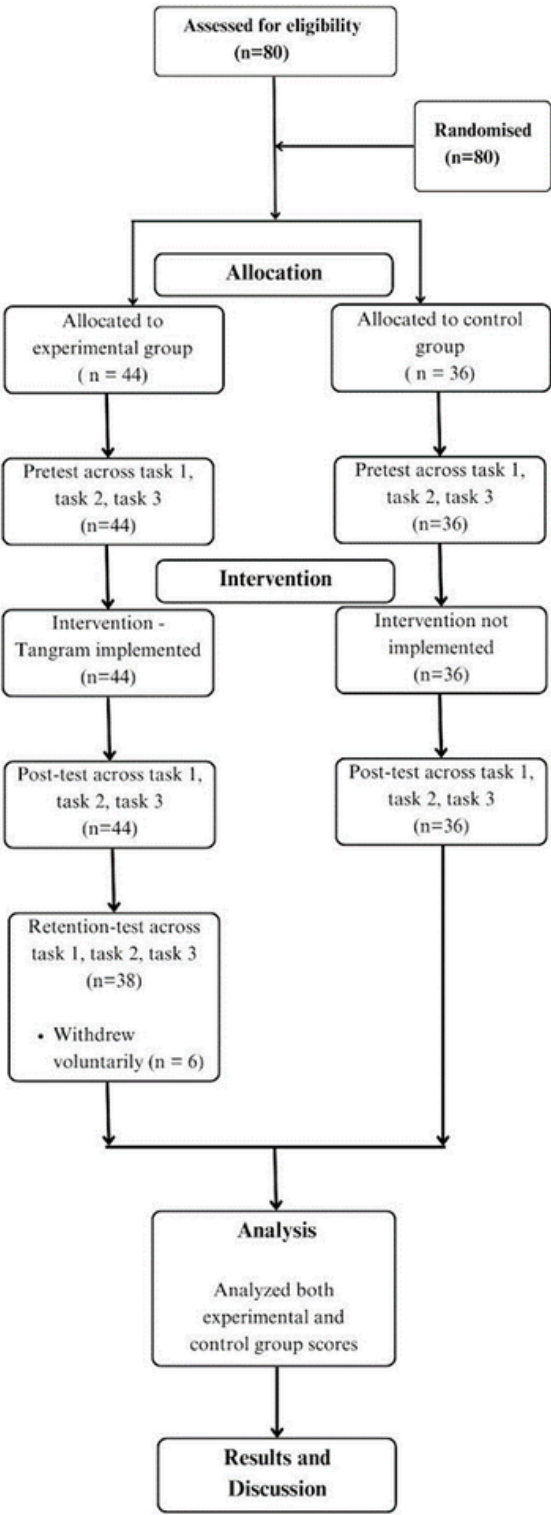


Figure 1: Shows the flowchart of the experimental design

Variables:

The variables of the study were as follows:

Independent variable: The independent variable was the intervention game, which induced changes in the dependent variable. The Tangram game was administered to the participants for this variable.

Dependent variable: The dependent variables included spatial-temporal reasoning, velocity, and precision.

Sample:

The population for the study were as follows:

Universal population : universal population consisted of school-aged children.

Target population: The target population consisted of school-aged children aged 6 to 11 years.

Inclusion criteria:

- Child of school age
- Between the ages of 6 and 11 years
- English-speaking children

Exclusion criteria:

The sample population excluded participants who met these criteria:

- Children exhibiting developmental delays
- Children residing in hostels or other facilities away from their parents.

Procedure:

Participants were chosen according to established inclusion and exclusion criteria and were randomly assigned to either the experimental group or the control group. Consent was acquired from the parents and educators. The participants were first requested to provide fundamental demographic information, followed by an explanation of the study's objectives. Thereafter, following To establish rapport, they underwent three tasks designed to evaluate spatial-temporal reasoning as a pre-test.

The three tasks were:

Task1: Pattern Recognition and Completion: Exhibiting a pattern or sequence that is partially finished. The child was instructed to complete the pattern or series by incorporating the absent elements. This activity assessed a young person's ability to recognize and anticipate spatial patterns.

Task 2: The student was assigned a maze puzzle for navigation. The child was instructed to navigate from one end of the maze to the opposite end. This assessed the individuals' spatial planning and execution abilities.

Task 3: Jigsaw puzzles: The child was instructed to assemble the puzzle pieces to form the complete image. This facilitated the assessment of their ability to manipulate puzzle pieces and conduct spatial analysis.

The velocity and precision were evaluated for the three tasks. Subsequently, the participants in the Experimental group underwent the tangram game intervention. A post-test was administered for the identical tasks following the training period. The control group did not participate in the intervention game; instead, they were allotted a 15-minute break during which they could engage in drawing on paper to occupy themselves. A post-test was administered following the break. The data was evaluated, analyzed, and statistically processed to derive results using Jamovi software. To assess the retention of the intervention over an extended period, participants were re-evaluated on the three tasks after 10 days.

Intervention game: A tangram is a puzzle comprising seven flat polygons, known as "tans," which can be assembled to form various shapes. The objective of the game is to replicate a design, presented solely as an outline and commonly located in a puzzle book, utilizing all seven pieces without any overlap. Tangrams are extensively utilized to instruct various mathematical concepts, and including congruence, symmetry, geometry, and perimeter, in addition to enhancing perceptual reasoning, visual-spatial awareness, and creativity

Statistical analysis :

The analysis was conducted utilizing Jamovi software. Descriptive statistics were conducted on the data to comprehend the variations in time taken, accuracy, and differences across various tasks. The pre-test and post-test scores were compared through the mean and standard deviation, utilizing independent sample tests for analysis. A paired sample t-test was employed to compare the pre-test and post-test scores of both groups to assess the changes

in time taken and accuracy rate following the intervention. An ANOVA was performed to compare the Pre-test, Post-test, and retention test groups in order to assess the variance among the groups and evaluate the efficacy of the tangram intervention. Subsequent to this, a Post HOC test was conducted to analyze pairwise comparisons among the groups and to elucidate the differences in values between the three groups.

Results and Discussion:

Results:

The primary aim of the study was to assess the efficacy of utilizing tangram as a means to improve spatial-temporal reasoning in children aged 6 to 11 years. A variety of statistical analysis methods were utilized to accomplish this objective. The study comprised 80 participants, with 44 randomly assigned to the experimental group and 36 to the control group. Thirty-eight participants were present for the retention test of the experimental group. The data did not conform to a normal distribution according to the normality tests conducted.

The data results are as follows:

Independent- sample-t- test

H01a: There exists no significant disparity between the Pre-test control group and the Pre-test experimental group regarding task 1, task 2, and task 3.

Task	Dimensions	Pre-test Control group		Pre-test Experimental group		P - value
		M1	SD1	M2	SD2	
Task 1	Time (sec)	29	13.7	36.27	13.271	0.017
	Accuracy	1.36	0.543	1.55	1.109	0.375
Task 2	Time (sec)	56.97	28.084	58.11	19.363	0.23
	Accuracy	2.39	1.153	2.66	1.524	0.413
Task 3	Time (sec)	228.89	61.318	287.59	75.898	0.001
	Accuracy	5.58	1.273	6.45	1.284	0.006

Table 1: Indicating independent sample-t-test of Pre-test control and Pre-test experimental group across task 1, task 2 and task 3

The table above contrasts the performance of students in the experimental and control groups across three distinct tasks prior to the implementation of the intervention. In task 1, the experimental group (M2 = 36.27 sec) exhibited a significantly longer duration than the control group (M1 = 29 sec), t (2.398). The experimental group exhibited marginally lower accuracy than the control group; however, the difference was not statistically significant.

In task 2, the control group, M1 = 1.36, required a duration nearly equivalent to that of the experimental group, M2 = 1.55, as the values exhibit minimal disparity. Concerning accuracy, no substantial difference exists between the experimental group (M2 = 2.66) and the control group (M1 = 2.39). In task 3, the control group (M1 = 228.89) exhibited a shorter duration than the experimental group (M2 = 287.59). The accuracy was found to be nearly equivalent in the experimental group (M2 = 6.45) and the control group (M1 = 5.58).

H01b: There is no significant difference between the post-test control group and the post-test experimental group for tasks 1, 2, and 3.

Task	Dimensions	Post-test Control group		Post-test Experimental group		P - value
		M1	SD1	M2	SD2	
Task 1	Time (sec)	24.61	10.782	21.95	10.781	0.224
	Accuracy	1.33	0.717	1	0.778	0.095
Task 2	Time (sec)	46.86	19.221	37.89	12.114	0.035
	Accuracy	1.92	1.18	0.5	0.665	0.001
Task 3	Time (sec)	212.69	50.945	213.84	52.92	0.831
	Accuracy	5.08	1.36	1.77	1.054	0.001

Table 2: Indicating independent sample-t-test of Post-test Control and Post-test Experimental Group across task 1, task 2 and task 3

The table contrasts the post-test control group with the post-test experimental group. The post-test experimental group (M2 = 21.95) completed task 1 in less time than the post-test control group (M1 = 24.6), indicating a slight difference in duration. The accuracy is superior in the post-test experimental group (M2 = 1.00) compared to the post-test control group (M1 = 1.33). In task 2, the duration required for the experimental group (M2 = 37.89) to complete the task is significantly less than that of the control group (M1 = 46.86) in the post-test. The accuracy in the control group (M1 = 1.92) is inferior to that of the experimental group (M2 = 0.665). The post-test control group and experimental group exhibit minimal significant differences regarding the duration required for task 3. The accuracy in the post-test control group (M1 = 5.08) is markedly inferior to that of the post-test experimental group (M2 = 1.77). There is minimal difference in time between tasks 1 and 3; however, there is a substantial difference in both time and accuracy in task 2, as well as in accuracy in task 3. The p-value is 0.001, which is less than 0.05; therefore, we can reject the null hypothesis, concluding that a difference exists between the post-test control group and the post-test experimental group.

Paired sample-t- test:

H01c: There is no substantial difference between the pre-test experimental group & the post-test experimental group.

Task	Dimensions	Pre-test experimental		Post-test Experimental group		Sig. value
		M1	SD1	M2	SD2	
Task 1	Time (sec)	36.27	13.271	21.95	10.781	0.001
	Accuracy	1.55	1.109	1	0.778	0.001
Task 2	Time (sec)	58.11	19.363	37.89	12.114	0.001
	Accuracy	2.66	1.524	0.5	0.665	0.001
Task 3	Time (sec)	287.59	75.898	213.84	52.92	0.001
	Accuracy	6.45	1.284	1.77	1.54	0.001

Table 3: Indicating paired Sample-t-test of Pre-test experimental and Post-test Experimental across task 1, task 2 and task 3

The table compares the pre-test and post-test experimental groups using paired sample t-test metrics. A notable reduction in the time required for task 1 was observed in the experimental group post-intervention (M1 = 36.27; M2 = 21.95). The accuracy has marginally improved in the post-experimental group. In task 2, the duration recorded by the post-test group is markedly less than that of the pre-test group. The mean for the pre-test group is M1 = 58.11 and for the post-test group is M2 = 37.89, while the number of errors in the post-test group has diminished from M1 = 2.66 to M2 = 0.50. The duration has markedly diminished in the post-test group (M2 = 213.84) relative to the pre-test group (M1 = 287.59). The accuracy in the post-test group (M2 = 1.77) has also improved relative to the pre-test group (M1 = 6.45). The p-value is less than 0.05 (p = 0.001) across all three tasks; therefore, we reject the null hypothesis and conclude that a significant difference exists between the pre-test experimental & post-test experimental groups.

H01d: There is no substantial difference between the pre-test control and post-test control for tasks 1, 2, and 3.

Task	Dimensions	Pre-test control		Post-test control group		P - value
		M1	SD1	M2	SD2	
Task 1	Time (sec)	29	13.77	24.61	10.782	0.009
	Accuracy	1.36	0.543	1.33	0.717	0.821
Task 2	Time (sec)	56.97	28.084	46.86	19.221	0.002
	Accuracy	2.39	1.153	1.92	1.18	0.025
Task 3	Time (sec)	228.89	61.318	213.84	52.92	0.003
	Accuracy	5.58	1.273	5.08	1.36	0.074

Table 4: Indicating paired sample-t-test of Pre-test control and Post-test Control Group across task 1, task 2 and task 3

The table above contrasts the pre-test control group with the post-test control group. The table in task 1 indicates no substantial difference between the means for time taken (M1 = 29; M2 = 24.61) and accuracy (M1 = 1.36; M2 = 1.33) across both groups. In task 2, the post-test control group exhibited a marginal reduction in time (M2 = 24.61) relative to the pre-test control group (M1 = 56.97). There is no distinction between the means of the pre-test control group (M1 = 2.39) and the post-test control group (M2 = 1.92). The duration has marginally diminished in the post-test control group (M2 = 213.84) relative to the pre-test control group (M1 = 228.89) for task 3. There is no significant difference in accuracy between the pre-test control group (M1 = 5.58) and the post-test control group. M2 equals 5.08 The table indicates that, with the exception of accuracy in task 1 ($p > 0.05$, $p = 0.821$) and task 3 ($p > 0.05$, $p = 0.074$), the significance values for time in all three tasks and accuracy in task 2 are below 0.05. Since $p < 0.05$, we partially reject the null hypothesis. Consequently, we can ascertain that a substantial disparity exists between the pre-test control group and the post-test control group.

H02: There is no significant difference among the Pre-test Experimental group, the Post-test Experimental group, and the Retention Test for tasks 1, 2, and 3.

Repeated measures ANOVA:

Task	Dimensions	Pre-test experimental	Post-test Experimental	Retention test	F	Mean squares
		M1	M2	M3		
Task 1	Time (sec)	36.27	21.95	25.32	16.53	2195.72
	Accuracy	1.55	1	1.11	3.99	3.487
Task 2	Time (sec)	58.11	37.89	40.68	25.397	5296.629
	Accuracy	2.66	0.5	0.86	37.595	46.173
Task 3	Time (sec)	287.59	213.84	218.81	20.282	80419.11
	Accuracy	6.45	1.77	2	220.7	252.934

The aforementioned tables utilize Repeated Measures ANOVA to compare pre-test experimental, post-test experimental, & retention test groups. The variance among the means of the three groups for the duration of task 1 is $F = 16.530$. In the case of Accuracy, the variation between the groups is $F = 3.990$. For Task 2, the mean variation in time among the three groups is 25.397, while the variation in accuracy is 37.595. The difference in the means of the three groups for the time taken in task 3 is 20.282, while the difference in accuracy among the groups is 220.70.

POST HOC's TEST :

Task	Dimensions	Pre-test experimental	Post-test Experimental group	Retention test	Sig. value
		M1	M2	M3	
Task 1	Time (sec)	36.27	21.95	25.32	0.365
	Accuracy	1.55	1	1.11	0.856

Table 6: Indicating Post Hoc Test for Time and Accuracy in Task 1 across Pre-test Experimental, Post-test Experimental and retention test experimental group

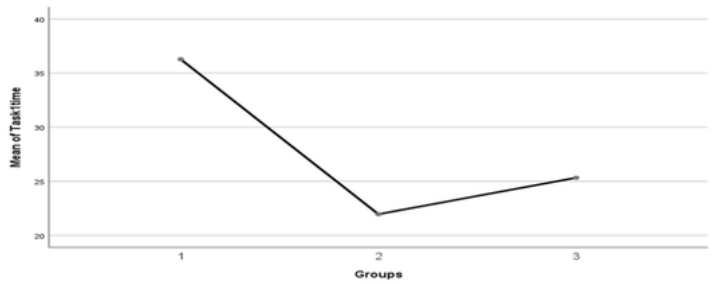


Figure 2: Indicating mean of Task 1 time across the three groups, Pre-test experimental, Post-test experimental and retention test

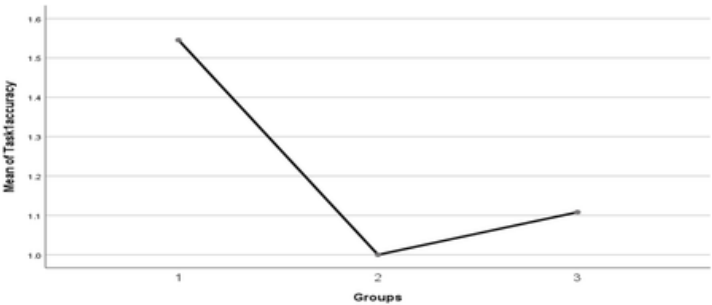


Figure 3: Indicating mean of Task 1 accuracy across the three groups, Pre-test experimental, Post-test experimental and retention test

POST HOC's TEST:

The data in Task 1 indicates that the post-test experimental group (M2 = 21.95) and the retention test group (M3 = 25.32) require less time than the pre-test experimental group (M1 = 36.27). The accuracy has marginally improved in the post-test experimental group (M2 = 1.00) and retention test group (M3 = 1.11) relative to the pre-test experimental group (M1 = 1.55). Figures 1 and 2 illustrate that there is no significant difference in the means of time and accuracy between the Post-test and retention test for task 1, while a significant difference exists between the Pre-test and the other two groups. This indicates a substantial reduction in the time required to complete task 1, accompanied by an increase in accuracy within the post-test and retention test groups.

Task	Dimensions	Pre-test experimental	Post-test Experimental group	Retention test	Sig. value
		M1	M2	M3	
Task 2	Time (sec)	58.11	37.89	40.68	0.661
	Accuracy	2.66	0.5	0.86	0.28

Table 7: Indicating Post Hoc’s Test for Time & Accuracy in Task 2 across Pre-test experimental, Post-test experimental and retention test experimental group

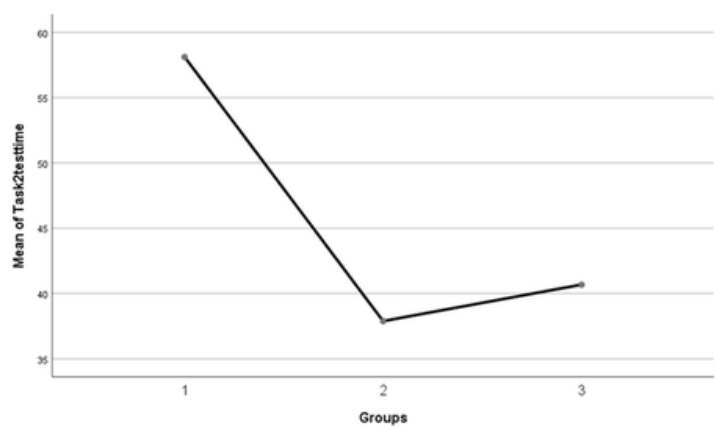


Figure 4: Indicating mean of Task 2 time across the three groups, Pre-test experimental, Post-test experimental and retention test

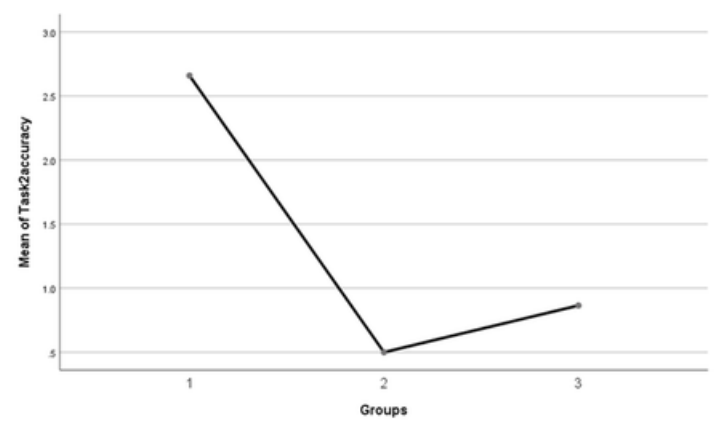


Figure 5. Indicating mean of Task 2 accuracy across the three groups, Pre-test experimental, Post-test experimental and retention test

The table indicates a substantial disparity in the duration required for task 2 across the three groups. The post-test experimental group (M2 = 37.89) and retention group (40.68) required less time than the pre-test experimental group (58.11). The accuracy among participants has improved following the intervention in the post-test experimental group (M2 = 0.50) and the retention experimental group (M3 = 0.86) compared to the pre-test experimental group (M1 = 2.66). Figures 3 and 4 indicate a minor disparity in the means of time and accuracy between the Post-test and retention test for task 2. Upon comparing the pre-test, post-test, and retention test groups, it is evident that the duration required to complete task 2 has diminished following the intervention, while accuracy has markedly improved post-implementation.

POST HOC’s TEST:

Task	Dimensions	Pre-test experimental	Post-test Experimental group	Retention test	Sig. value
		M1	M2	M3	
Task 3	Time (sec)	287.59	213.84	218.81	0.934
	Accuracy	6.45	1.77	2	0.61

Table 8: Indicating Post Hoc’s Test for Time and Accuracy in Task 3 across Pre-test experimental, Post-test experimental and retention test experimental group

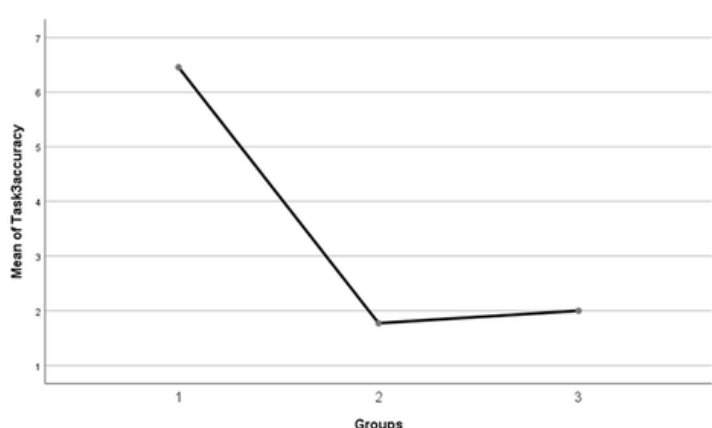


Figure 6: indicating comparison of mean of Task 3 accuracy across the three groups, Pre-test experimental, Post-test experimental and retention test

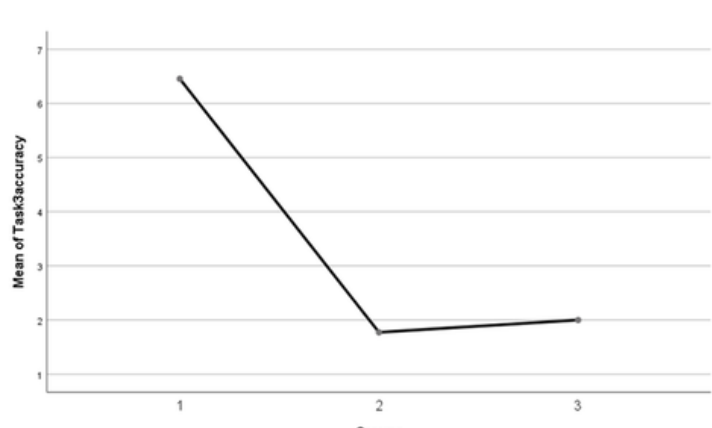


Figure 7. Indicating comparison of the mean of Task 3 accuracy across the three groups, Pre-test experimental, Post-test experimental and retention test

In task 3, the duration recorded by participants markedly diminished in the post-experimental group (M2 = 213.84) and retention test group (M3 = 218.81) relative to the pre-test experimental group (M1 = 287.59).

The accuracy rate in the post-test experimental group ($M2 = 1.77$) and the retention test experimental group ($M3 = 2.00$) shows a notable increase compared to the pre-test experimental group ($M1 = 6.45$).

Figures 1 & 2 illustrate that there is no significant difference in the means of time and accuracy between the Post-test and retention test for task 3, while a significant difference exists between the Pre-test and the other two groups. This indicates that the duration required to finish task 3 has considerably decreased, while accuracy has improved in both the post-test and retention test groups following the intervention. Since $p > 0.05$, we reject the null hypothesis; thus, we conclude that a significant difference exists among the post-test experimental group and the retention test experimental group. It is evident that there is no substantial difference between the post-test and retention test, while both groups exhibit significant variation from the pre-test control group. Therefore, it can be concluded that the intervention improved the participants' speed and accuracy. The figures illustrate the post-HOC test values in the schematic representation below:

Discussion:

The findings of the experimental study offer significant insights into the efficacy of the intervention on participants' performance across three tasks: the pattern completion task, the labyrinth task, and the jigsaw puzzle task. The initial comparison between the pre-test control and pre-test experimental groups revealed a minor discrepancy in performance, with the experimental group taking marginally longer on task 1; however, the accuracy rate and time for the other tasks remained consistent. Therefore, it is concluded that no significant difference exists between the pre-test control group and the pre-test experimental group. The post-test comparison between the control and experimental groups revealed substantial enhancements in both time and accuracy, especially in tasks 2 and 3 within the experimental group. The paired sample test comparison between the pre-test and post-test experimental groups demonstrated a significant improvement in both time and accuracy across all tasks following the intervention. The findings indicate a direct effect of the Tangram intervention on enhancing participant performance. Research indicates that tangram has a beneficial impact on enhancing visual reasoning (Ji, A., et al., 2022). With the exception of the accuracy in tasks 1 and 3 ($p > 0.05$, $p = 0.821$ and 0.074 , respectively), it is apparent that the significance values in the comparative analyses of the pre-test control group and post-test control group are below 0.05. Consequently, it is clear that the Pre-test control group & the Post-test control

group exhibit a difference ($M1 = 29.00$; $M2 = 24.61$). This may arise from the exposure effect and the practice effect. Research on the mere exposure effect has revealed that recognition memory may serve as a constraining factor that could affect the experimental outcomes in participants. (Bornstein & Craver-Lemley, 2004; Bornstein, 1989). The theory of perceptual fluency/attribution elucidates that prior exposure to a stimulus enhances the ease of processing that stimulus upon subsequent presentation (Lanska et al., 2014). This may elucidate the potential rationale for the successful identification and effective engagement in the tasks. The retention test included only 38 participants, as the other students withdrew from the study. The repeated measures ANOVA revealed significant differences among the pre-test, post-test, and retention-test experimental groups, indicating the enduring effect of the Tangram intervention on performance. The Post HOC test further corroborated these findings, demonstrating a consistent enhancement in both time and accuracy across all tasks. Limited research suggests that Tangram enhances students' higher-order thinking and problem-solving strategies, with effects persisting over an extended period (Lin, C., et al., 2022). These findings demonstrate the intervention's efficacy in improving participants' performance across tasks, indicating that the Tangram intervention significantly enhances spatiotemporal reasoning in participants.

Summary and Conclusion:

The research concluded that children aged 6 to 11 can enhance their spatial and temporal reasoning skills through the use of tangrams. Eighty individuals participated in the study, comprising students from a school in Bangalore. Statistical analyses, including the independent t-test, paired sample t-test, and ANOVA, were employed to compare the pre-test, post-test, and retention tests in order to evaluate the effectiveness of the tangram as an intervention. The results indicated that, subsequent to the intervention, the experimental group's time & accuracy markedly enhanced. Moreover, the advantageous effects of the tangram seem to endure over time. The study's findings indicate that children can enhance their spatial-temporal reasoning skills through the use of tangrams. Cognitive abilities of students can be enhanced in educational environments through the implementation of such interventions. This study underscores the significance of game-based interventions for cognitive skills, thereby creating avenues for further exploration into the efficacy of alternative game-based interventions. The study's findings are significant for informing education policy to improve diverse cognitive skills in children. Policymakers can be incentivized to endorse the incorporation of game-based interventions in

educational policies and initiatives. Further research is advised to ascertain the long-term effects of tangram interventions and other game-based strategies that may augment children's cognitive development. By financing analogous programs, legislators can enhance children's cognitive development within the community, thereby impacting their academic success.

Implications:

The study's findings have implications for the development and implementation of various interventions designed to enhance cognitive performance and abilities. The study offers significant insights into the efficacy of the tangram intervention in improving spatial-temporal reasoning in children aged 6 to 11 years. These interventions can also facilitate the promotion of government education policy formulation. Incorporating such Incorporating games into classroom environments can benefit both typical learners and those with special needs. Tangram is readily accessible and understandable, allowing for its use without intricate explanations, enabling parents to effectively engage their children.

Limitations:

Notwithstanding the favorable outcomes, there are aspects of the study that can be enhanced. It is essential to recognize the possible constraints of the sample size and the randomization of participants in the study. Another issue to consider is the influence of confounding variables, including students' motivation levels, prior experience, and external factors impacting task performance. Furthermore, employing additional well-researched tasks can improve the accuracy of effect measurement; thus, subsequent studies should contemplate integrating diverse measures for a more definitive evaluation.

Recommendation:

Tangram can be integrated into preschool and elementary school curricula. Likewise, through such activities, educators can cultivate an environment that emphasizes the development of diverse cognitive skills from an early age. Consequently, additional research should concentrate on investigating further activities or interventions that would facilitate the enhancement of diverse cognitive abilities. Moreover, the influence of the tangram intervention on long-term retention warrants investigation in subsequent studies. Future research may investigate participants' performance over an extended period and the sustainability of the intervention's effectiveness in improving spatial-temporal reasoning among participants.

It is essential to investigate supplementary factors that may affect cognitive development. Therefore creating customized interventions to address the unique needs of individual children can facilitate the design of targeted strategies for enhancement, potentially leading to improved learning outcomes.

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